

Operations Research

Linear Programming III: Simplex algorithm

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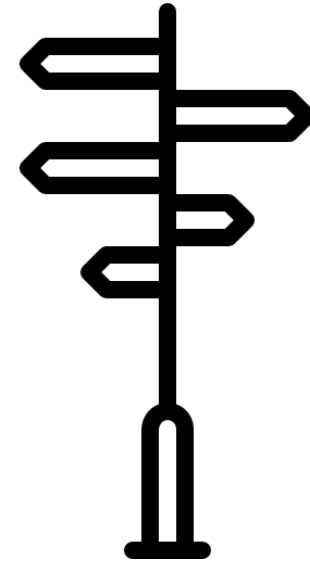
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Course outline

- **Linear Optimization**
 - Modeling with linear programming and graphical solution
 - Solving linear programs, convexity
 - **Simplex algorithm, 2-phase Simplex**
 - Simplex algorithm in matrix notation
 - Sensitivity analysis, re-optimization
 - Duality theory, zero-sum games
- **Integer Linear Optimization**
 - Modeling integer programming problems (IPs)
 - Solving IPs: branch & bound, cutting planes
 - Advanced solution methods: column generation, approximation algorithms
 - Fast solvable integer problems: total unimodularity, matroids
 - Network flow problems
 - Traveling salesperson problem
- **Nonlinear Optimization**
 - Optimization of multidimensional functions
 - Convex optimization



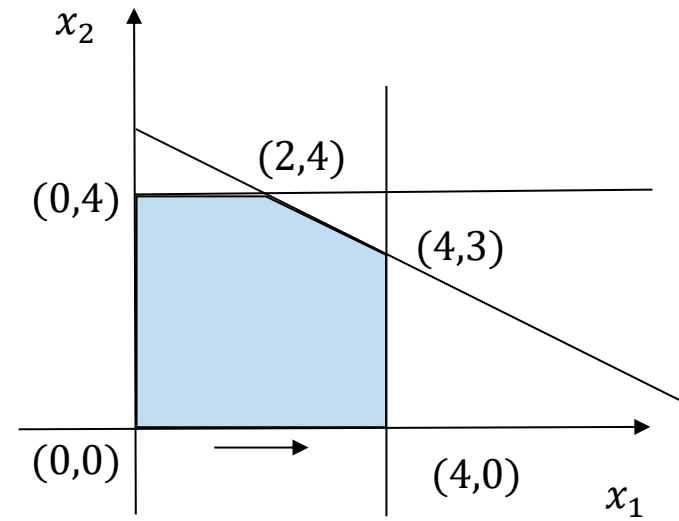
Today you will learn ...

- the algorithmic description of the **Simplex algorithm**
- how to solve linear programs with the **Simplex tableau**
- how to find **initial feasible solutions** of LPs



Recap

$$\begin{array}{llllll} \max & 2x_1 & + & x_2 & & \\ \text{s.t.} & x_1 & + & 2x_2 & \leq & 10 \\ & x_1 & & & \leq & 4 \\ & & & x_2 & \leq & 4 \\ & & & & x_1, x_2 & \geq 0 \end{array}$$



- solutions are in the corners (**Basic solutions**)
- an optimal solution can be found by iteratively visiting adjacent corners

→ Solution of systems of equations

Recap

After each Simplex iteration, the system of equations should be converted to **canonical form**.

Canonical form:

- standard form [= - conditions]
- positive right hand side
- each equation contains a variable with coefficient 1, which has coefficients of 0 in all the other equations.

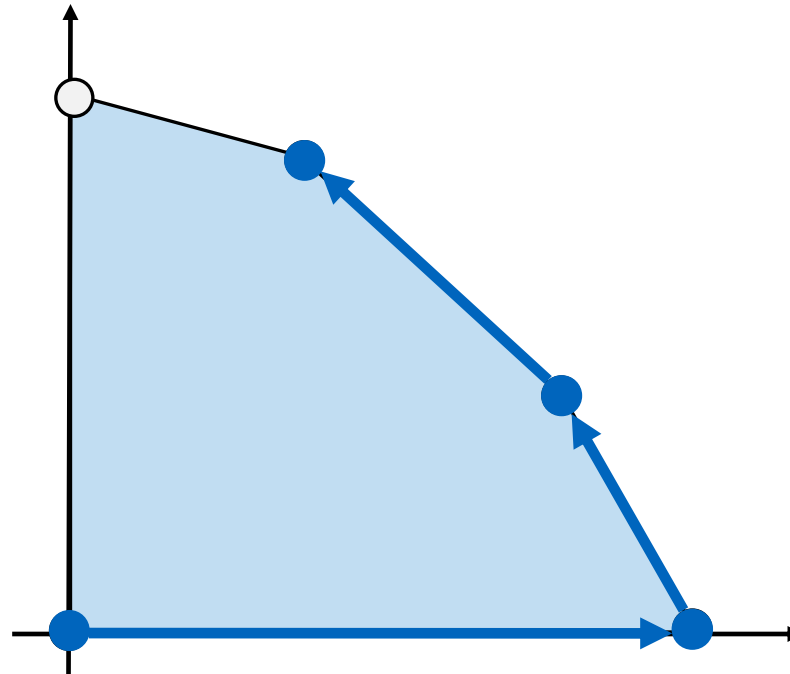
In canonical form, it is easy to calculate how the basic variables and z change when a non-basic variable is increased.

The transformation into canonical form can be done via row operations.

Simplex

Simplex. (Dantzig 1947)

- developed after WW2 for logistic problems
- practical solution method that searches adjacent extreme points.
- exponential "worst-case" time complexity, polynomial implementations are not known (Klee&Minty, 1972).



Pseudocode Simplex Algorithm

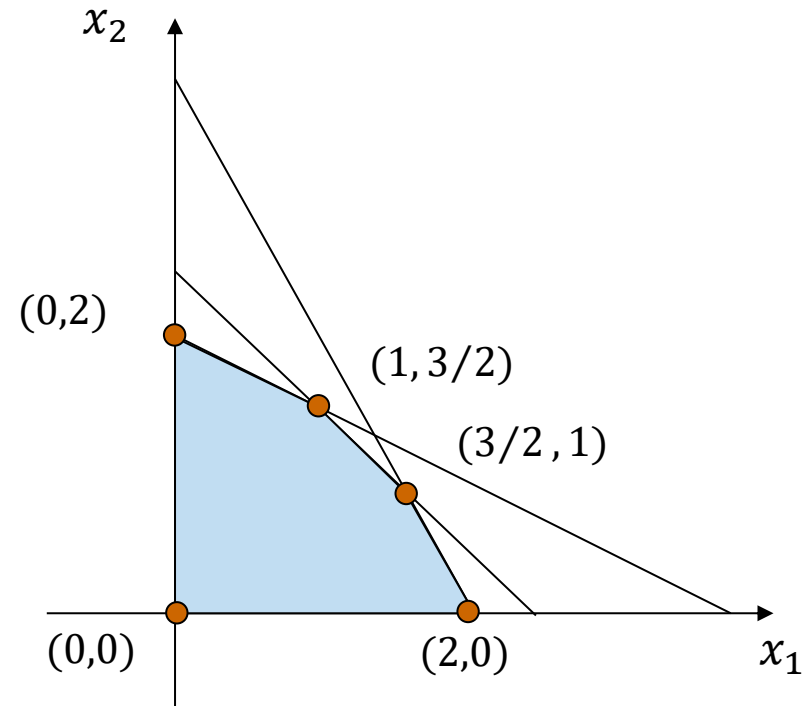
- **Find an initial basis** and formulate the LP in canonical form for that basis.
- Repeat {
 - **Find a new basic variable** by analyzing the first equation with z in canonical form.
 - **Find an outgoing basic variable** - if none can be found, then the LP is unbounded.
 - Convert the system of equations into a **new canonical form** where the new basic variable is only present in the equation where the outgoing basic variable is currently present.
- } Until there is no more non-basic variable that can be increased to increase z .

Another example

$$\begin{array}{llllll}
 \max & x_1 & + & x_2 & = & z \\
 \text{s.t.} & 2x_1 & + & x_2 & \leq & 4 \\
 & x_1 & + & 2x_2 & \leq & 4 \\
 & 2x_1 & + & 2x_2 & \leq & 5 \\
 & & & x_1, x_2 & \geq & 0
 \end{array}$$

Transform into:

$$\begin{array}{llllll}
 \max & z & & & & \\
 z & - & x_1 & - & x_2 & = & 0 \\
 & 2x_1 & + & x_2 & + x_3 & = & 4 \\
 & x_1 & + & 2x_2 & & + x_4 & = & 4 \\
 & 2x_1 & + & 2x_2 & & & + x_5 & = & 5 \\
 & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
 \end{array}$$



Choose x_1, x_2 as
non-basic variables
 $\rightarrow BFS = (0,0,4,4,5)$
 $\rightarrow z = 0$

Example

$$\begin{array}{rclcl}
 z & -x_1 & -x_2 & & = 0 \\
 & 2x_1 & +x_2 & +x_3 & = 4 \\
 & x_1 & +2x_2 & & +x_4 = 4 \\
 & 2x_1 & +2x_2 & & +x_5 = 5 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{array}$$

The value of z can be maximized further if x_1 is increased.

A simplification for calculation... the Simplex tableau

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1	-1	-1				0
x_3		2	1	1			4
x_4		1	2		1		4
x_5		2	2			1	5

Example

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1	-1	-1				0
x_3		2	1	1			4
x_4		1	2		1		4
x_5		2	2			1	5

By how much can we increase x_1 ?

Which basic variable becomes the non-basic variable?

For equation x_3 , x_1 can be increased to $4/2 \rightarrow 2$

For equation x_4 , x_1 can be increased to $4/1 \rightarrow 4$

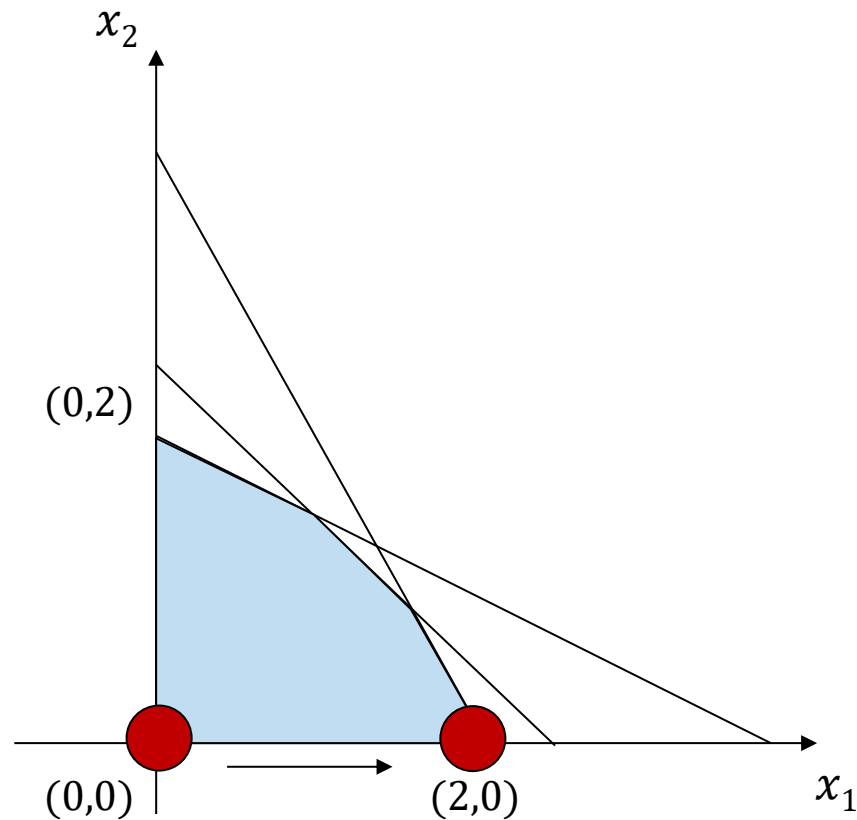
For equation x_5 , x_1 can be increased to $5/2 \rightarrow 2.5$

The equation with the smallest restriction is chosen
(in which the entering BV has a positive coefficient).

x_3 becomes the **exiting** and x_1 the **entering** basic variable.

Graphic solution

In the original 2-variable problem, the changes have the following consequence:



Example

Transformation into canonical form:

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1	-1	-1				0
x_1		2	1	1			4
x_4		1	2		1		4
x_5		2	2			1	5



Divide by 2

Example

Transformation into canonical form:

Basic variable	Coefficients						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1	-1	-1				0
x_1		1	0.5	0.5			2
x_4		1	2		1		4
x_5		2	2			1	5





Add row x_1

Add $(-1) \cdot \text{row } x_1$

Add $(-2) \cdot \text{row } x_1$

Example

Transformation into canonical form:

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1		-0.5	0.5			2
x_1		1	0.5	0.5			2
x_4			1.5	-0.5	1		2
x_5			1	-1		1	1

Can z be improved more?

x_2 can still be increased!

Example

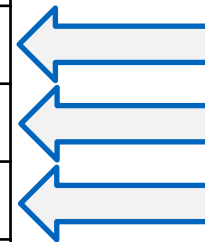
Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1		-0.5	0.5			2
x_1		1	0.5	0.5			2
x_4			1.5	-0.5	1		2
x_5			1	-1		1	1

We can increase x_2 by $1/1$ (smallest restriction in row x_5 !)

x_5 becomes the **exiting** basic variable and x_2 becomes the **entering** basic variable.

Example

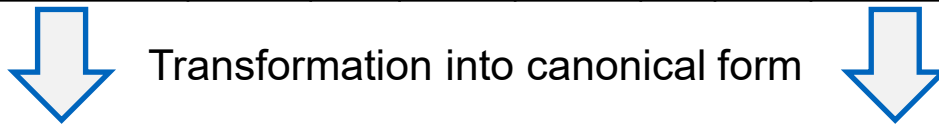
Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1		-0.5	0.5			2
x_1		1	0.5	0.5			2
x_4			1.5	-0.5	1		2
x_2			1	-1		1	1



Add $0.5 \cdot \text{row } x_2$

Add $(-0.5) \cdot \text{row } x_2$

Add $(-1.5) \cdot \text{row } x_2$

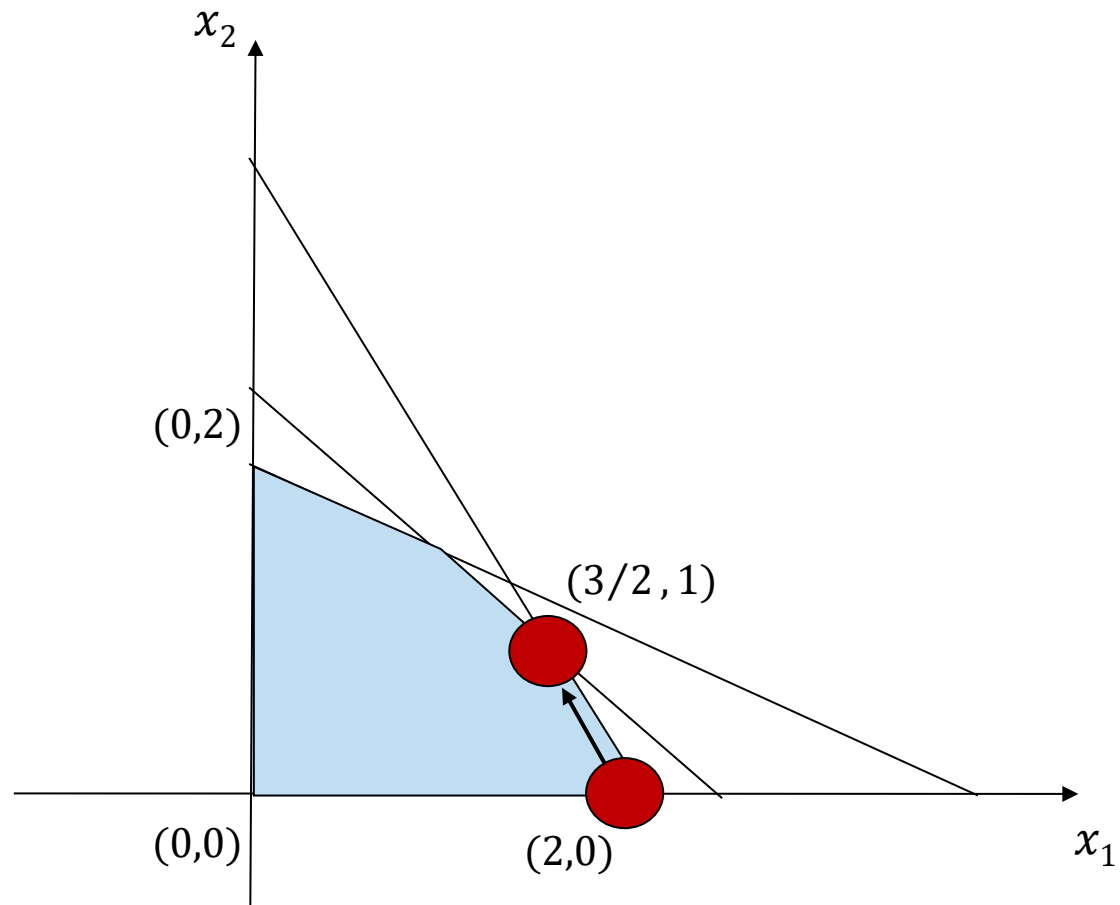


Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1					0.5	2.5
x_1		1		1		-0.5	1.5
x_4				1	1	-1.5	0.5
x_2			1	-1		1	1

Can z be improved more?

Is the solution optimal?

Graphic solution



Example

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1					0.5	2.5
x_1		1		1		-0.5	1.5
x_4				1	1	-1.5	0.5
x_2			1	-1		1	1



Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1					0.5	2.5
x_1		1			-1	1	1
x_3				1	1	-1.5	0.5
x_2			1		1	-0.5	1.5

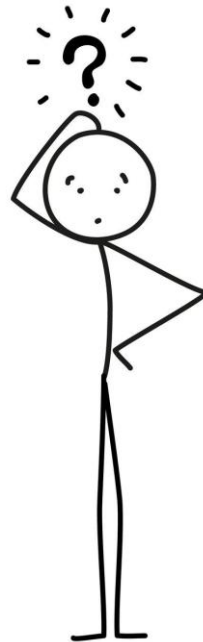
x_3 no longer appears in the z -line, even though x_3 is a non-basic variable.

What would happen, if we make x_3 a new basic variable?

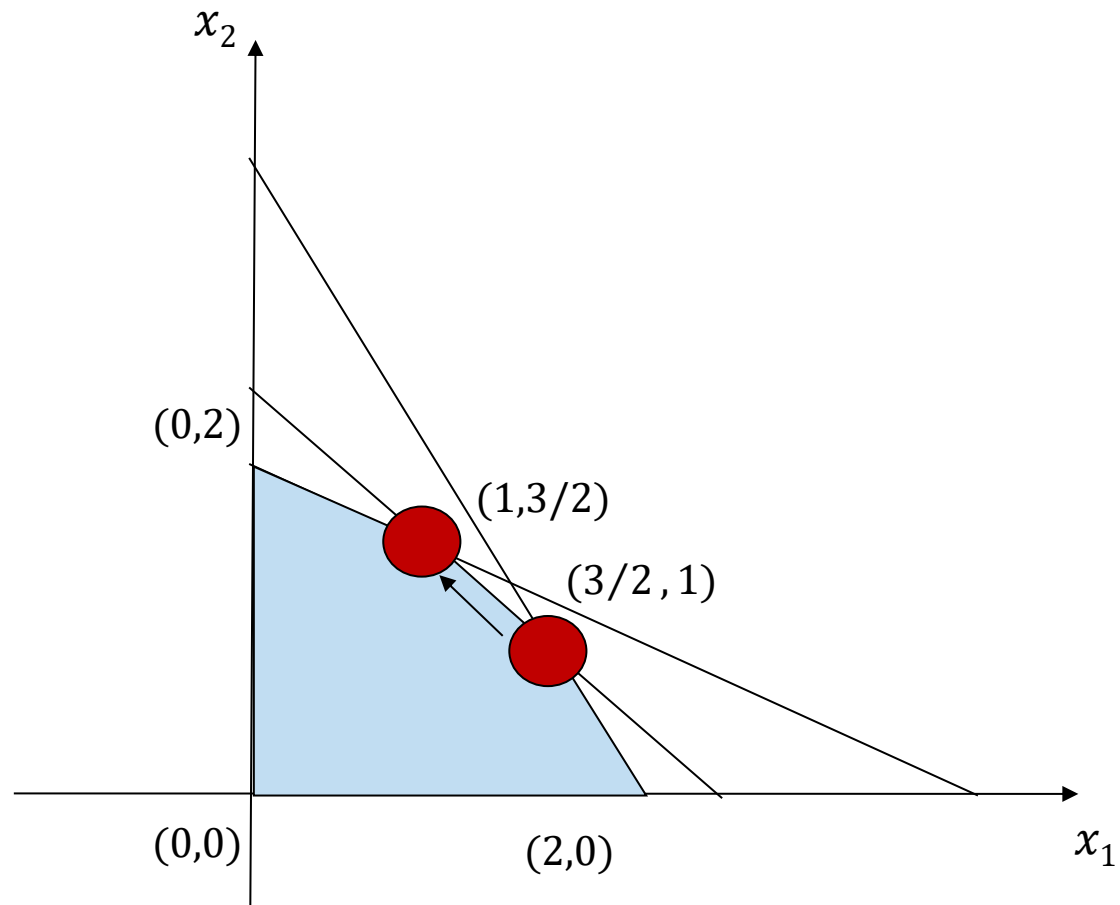
Example: Exchange with x_4

The target value z would not change with the coefficient 0, but it would be a new solution that is equally optimal.

Why does the objective function value not change
at the last basis exchange?



Graphic solution



Unboundedness

$$\begin{array}{ll} \max & x_1 \\ \text{s. t.} & x_1 - x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{array}$$

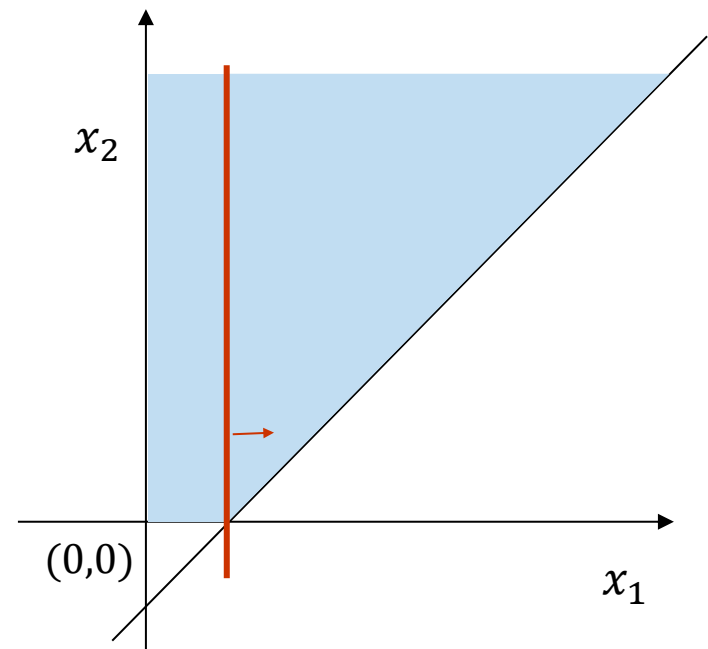
Basic variable	Coefficient				Solution
	z	x_1	x_2	x_3	
z	1	-1			0
x_3		1	-1	1	2



x_1 enters, x_3 leaves



Basic variable	Coefficient				Solution
	z	x_1	x_2	x_3	
z	1		-1	1	2
x_1		1	-1	1	2



Pseudocode Simplex Algorithm

- Find an initial basis and formulate the LP in canonical form for that basis.
- Repeat {
 - Find a new basic variable by analyzing the first equation with z in canonical form.
 - **Find an outgoing basic variable - if none can be found, then the LP is unbounded.**
 - Convert the system of equations into a new canonical form in which the new basic variable is present only in the equation in which the outgoing basic variable is currently present.
- } Until there is no more non-basic variable that can be increased to increase z .

Discussion

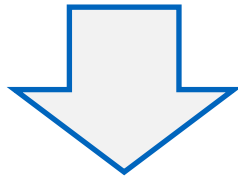
- Two possible new basic variables
 - If several non-basic variables are possible, so that each of them increases the target value, you can choose any of them.
 - You might want to choose the variable that increases the target value the most.
- New basic variable with value zero
 - If basic variables become zero due to the introduction of a new basic variable, problems can arise.

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1		-2	-1		2
x_1		1	1			2
x_4			3	1	1	6

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	2		-1		6
x_2		1	1			2
x_4		-3		1	1	0

Degeneration

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	2		-1		6
x_2		1	1			2
x_4		-3		1	1	0



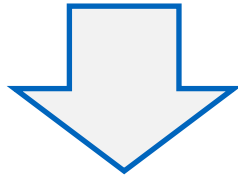
Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	-1			1	6
x_2		1	1			2
x_3		-3		1	1	0

An LP is degenerate if at least one basic variable has value 0.

- Graphical: More than 2 constraints intersect at one point.
- The objective function value remained the same. In this example, we can continue the simplex iteration.
- There are cases (not in this example) where the Simplex algorithm can cycle infinitely (rarely!).
- This can be prevented by using pivot rules when selecting the entering and exiting basic variables.

Final tableau, for completeness

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	-1			1	6
x_2		1	1			2
x_3		-3		1	1	0



Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	0	1	0	1	8
x_1		1	1			2
x_3			3	1	1	6

Klee and Minty [1972] gave a longer example in which the simplex algorithm really does cycle.

So, without modifications, the Simplex would not be a finite algorithm.

There are several rules (e.g., Bland's rule) to avoid cycling, but the proofs that these rules lead to a finite algorithm are long (and beyond this class).

Pseudocode Simplex Algorithm

- **Find an initial basis** and formulate the LP in canonical form for that basis.
- Repeat {
 - Find a new basic variable by analyzing the first equation with z in canonical form.
 - Find an outgoing basic variable - if none can be found, then the LP is unbounded.
 - Convert the system of equations into a new canonical form in which the new basic variable is present only in the equation in which the outgoing basic variable is currently present.
- } Until there is no more non-basic variable that can be increased to increase z .

Find an initial basis

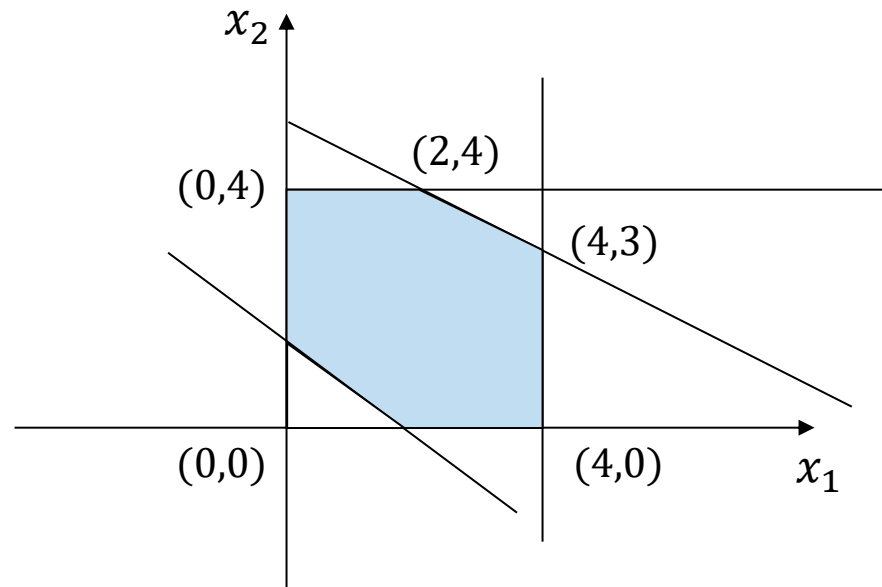
- How to find the initial set of basic variables?
- In the previous examples, we introduced slack variables for inequalities and declared the slack variables as initial bases.

$$\begin{array}{llllll}
 \max & 2x_1 & + & x_2 & & \\
 \text{s.t.} & 2x_1 & + & x_2 & \leq & 4 \\
 & x_1 & + & 2x_2 & \leq & 4 \\
 & 2x_1 & + & 2x_2 & \leq & 5 \\
 & & & x_1, x_2 & \geq & 0
 \end{array}$$

$$\begin{array}{llllllll}
 \max & 2x_1 & + & x_2 & & & & \\
 \text{s.t.} & 2x_1 & + & x_2 & + x_3 & & & = 4 \\
 & x_1 & + & 2x_2 & & + x_4 & & = 4 \\
 & 2x_1 & + & 2x_2 & & & + x_5 & = 5 \\
 & & & & & & x_1, \dots, x_5 & \geq 0
 \end{array}$$

Find an initial basis

Often the initial base is difficult to find or not feasible.

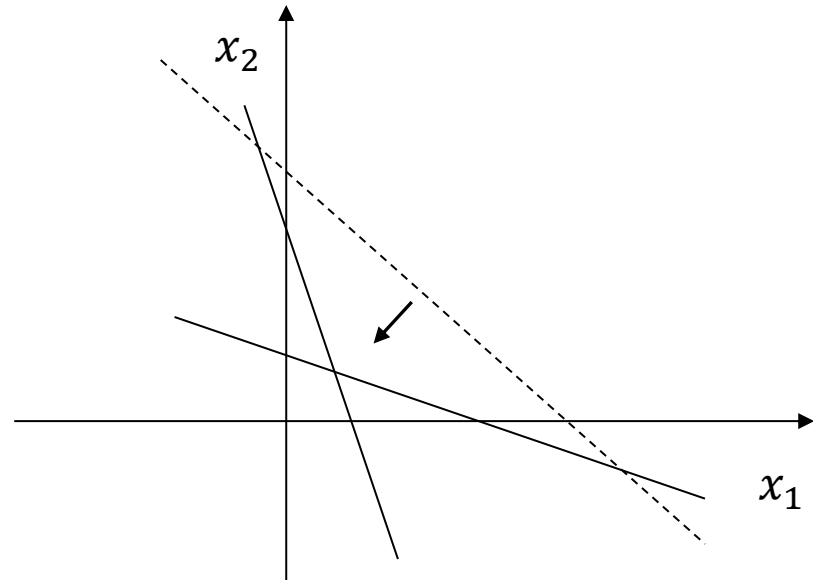


Find an initial basis

$$\begin{array}{llllll}
 \min & 2x_1 & + & 3x_2 & & \\
 \text{s.t.} & 1.7x_1 & + & 4.5x_2 & \geq & 35 \\
 & 4x_1 & + & x_2 & \geq & 35 \\
 & & & x_1, x_2 & \geq & 0
 \end{array}$$

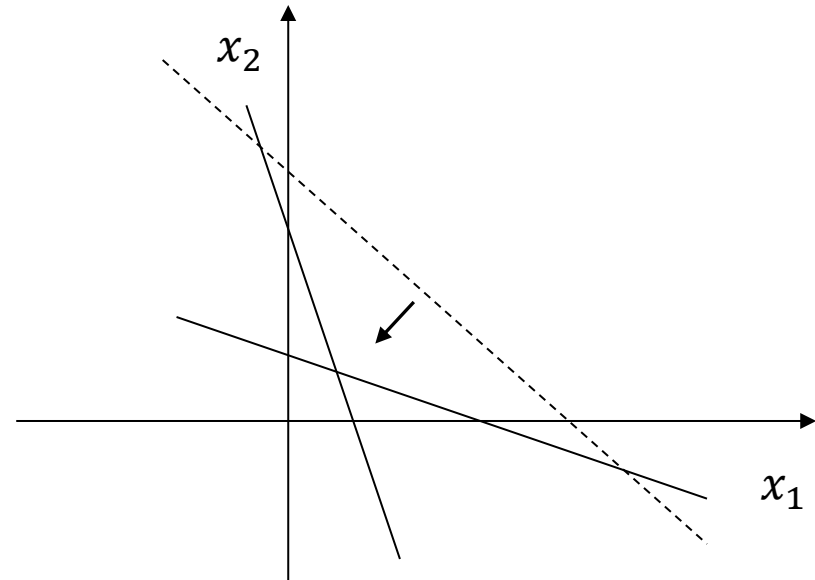
Reformulate to

$$\begin{array}{llllll}
 \max & -2x_1 & - & 3x_2 & & \\
 \text{s.t.} & -1.7x_1 & - & 4.5x_2 & \leq & -35 \\
 & -4x_1 & - & x_2 & \leq & -35 \\
 & & & x_1, x_2 & \geq & 0
 \end{array}$$



Find an initial basis

$$\begin{array}{llllll}
 \min & 2x_1 & + & 3x_2 & & \\
 \text{s.t.} & 1.7x_1 & + & 4.5x_2 & \geq & 35 \\
 & 4x_1 & + & x_2 & \geq & 35 \\
 & & & x_1, x_2 & \geq & 0
 \end{array}$$



Reformulate to

$$\begin{array}{llllllll}
 \max & z & & & & & & \\
 \text{s.t.} & z & + & 2x_1 & + & 3x_2 & & = & 0 \\
 & & - & 1.7x_1 & - & 4.5x_2 & + & x_3 & = & -35 \\
 & & - & 4x_1 & - & x_2 & & + & x_4 & = & -35 \\
 & & & & & & & & x_1, x_2, x_3, x_4 & \geq & 0
 \end{array}$$



$$\begin{array}{l}
 x_3 = -35 \\
 x_4 = -35
 \end{array}$$

Infeasible basis

Find an initial basis

If the LP is given in canonical form, we can easily find a BFS.

Example: LP in normal form with positive right side: introduce slack variables and select them as basic variables (the zero point is the first BFS).

This is equivalent to the zero point being part of the feasible region!

If this is not the case, you have to calculate a first BFS.

Two options:

- 2-phase method
- Big-M method

2-phase method

Setting: LP is not in canonical form.

That means basic variables (BV) for individual constraints (IC) are missing.

What now?

Create replacement problem (P')

- Introduce for each constraint j without basic variable an artificial variable w_j .
- $W = \{j \mid NB_j \text{ has no } BV\}$
- Constraints: $Ax + w' = b; x, w' \geq 0; w_j' = \begin{cases} w_j, & j \in W \\ 0, & \text{otherwise} \end{cases}$

The goal is to find a basic solution where w_j does not contribute.

$$(\sum_{j \in W} w_j = 0)$$

Example

$$\begin{array}{llllll}
 \max & -x_1 & - & 2x_2 & & \\
 \text{s.t.} & 2x_1 & + & x_2 & - & 4x_3 = -4 \\
 & 4x_1 & + & x_2 & + & 2x_3 = 4 \\
 & & & & & x_1, x_2, x_3 \geq 0
 \end{array}$$

$$\begin{array}{llllllll}
 \max & -x_1 & - & 2x_2 & & & & \\
 \text{s.t.} & -2x_1 & - & x_2 & + & 4x_3 & +w_4 & = 4 \\
 & 4x_1 & + & x_2 & + & 2x_3 & & +w_5 = 4 \\
 & & & & & & & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

The goal is to find an optimal solution for the new formulation,
where w_4 and $w_5 = 0$

This provides an initial basis for the original formulation!

Simplex Phase I

$$\begin{aligned} (P') \quad & \min \quad \sum_{j \in W} w_j = z_{new} \\ & s. t. \quad Ax + w' = b \\ & \quad \quad x, w' \geq 0 \end{aligned}$$

If $z_{new} = 0$:

The artificial variables do not impact feasibility.

It is possible to find a basis without w_j .

If $z_{new} > 0$:

Without the artificial variables, it is not possible to satisfy the constraints.

→ The feasible region is empty and the LP is infeasible.

Example

$$\begin{array}{ll}
 \max & -x_1 - 2x_2 \\
 \text{s.t.} & -2x_1 - x_2 + 4x_3 + w_4 = 4 \\
 & 4x_1 + x_2 + 2x_3 + w_5 = 4 \\
 & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \max & z_{\text{new}} \\
 \text{s.t.} & z_{\text{new}} + w_4 + w_5 = 0 \\
 & z + x_1 + 2x_2 = 0 \\
 & -2x_1 - x_2 + 4x_3 + w_4 = 4 \\
 & 4x_1 + x_2 + 2x_3 + w_5 = 4 \\
 & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

Problem: (P') not yet in canonical form.

Phase 1 Tableau

Basic variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1					1	1	0
z		1	1	2				0
w_4			-2	-1	4	1		4
w_5			4	1	2		1	4

→ Create canonical form

Basic Variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1		-2		-6			-8
z		1	1	2				0
w_4			-2	-1	4	1		4
w_5			4	1	2		1	4

Phase 1 Tableau

Basic variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1		-2		-6			-8
z		1	1	2				0
w_4			-2	-1	4	1		4
w_5			4	1	2		1	4

The old objective function z will be replaced in phase 1 by z_{new} .

Basic variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1		-5	-1.5		1.5		-2
z		1	1	2				0
x_3			-0.5	-0.25	1	0.25		1
w_5			5	1.5		-0.5	1	2

Phase 1 Tableau

Basic variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1		-5	-1.5		1.5		-2
z		1	1	2				0
x_3			-0.5	-0.25	1	0.25		1
w_5			5	1.5		-0.5	1	2

Basic variable	Coefficient							Solution
	z_{new}	z	x_1	x_2	x_3	w_4	w_5	
z_{new}	1					1	1	0
z		1		1.7		0.1	-0.2	-0.4
x_3				-0.1	1	0.2	0.1	1.2
x_1			1	0.3		-0.1	0.2	0.4

→ (0.4, 0, 1.2, 0, 0) is a basic feasible solution

Phase 2 Tableau

Eliminate the artificial slack variable and auxiliary objective function.

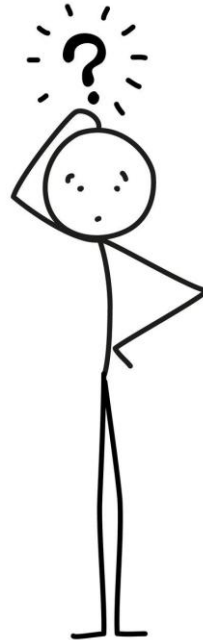
Basic variable	Coefficient				Solution
	z	x_1	x_2	x_3	
z	1		1.7		-0.4
x_3			-0.1	1	1.2
x_1		1	0.3		0.4

→ Apply standard Simplex

→ If the objective function value $z_{new} > 0$, then the original LP has no feasible solution

- 1.) How can you recognize that an LP is unbounded?
- 2.) How can you recognize that an LP is degenerate?

What can happen in such cases?



Find an initial basis

- 2-phase method
- Big-M method

Another possibility: Big- M method

Introduce slack variables with very high costs (number $-M$)

$$\begin{array}{llllllllll}
 \text{max} & z & & & & & & & & & \\
 \text{s.t.} & z & +x_1 & + & 2x_2 & & & +Mw_4 & +Mw_5 & = & 0 \\
 & & -2x_1 & - & x_2 & + & 4x_3 & +w_4 & & = & 4 \\
 & & 4x_1 & + & x_2 & + & 2x_3 & & +w_5 & = & 4 \\
 & & & & & & & & & & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

Claim:

Given a sufficiently large M and if the original problem is feasible, **there is an optimal basic feasible solution without artificial slack variables** in the basis.

Initial basis

The system with the artificial slack variables is in canonical form except for the first equation.

- The variables w_4 and w_5 still need to be eliminated from the equation.

$$\begin{array}{ll}
 \max & z \\
 \text{s.t.} & z + x_1 + 2x_2 + Mw_4 + Mw_5 = 0 \\
 & -2x_1 - x_2 + 4x_3 + w_4 = 4 \\
 & 4x_1 + x_2 + 2x_3 + w_5 = 4 \\
 & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \max & z \\
 \text{s.t.} & z + (1 - 2M)x_1 + 2x_2 - 6Mx_3 = -8M \\
 & -2x_1 - x_2 + 4x_3 + w_4 = 4 \\
 & 4x_1 + x_2 + 2x_3 + w_5 = 4 \\
 & x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{array}$$

Example

$$\begin{aligned}
 \max \quad & z \\
 \text{s.t.} \quad & z \quad (1 - 2M)x_1 \quad + \quad 2x_2 \quad - \quad 6Mx_3 \quad = \quad -8M \\
 & \quad \quad -2x_1 \quad - \quad x_2 \quad + \quad 4x_3 \quad + w_4 \quad = \quad 4 \\
 & \quad \quad 4x_1 \quad + \quad x_2 \quad + \quad 2x_3 \quad + w_5 \quad = \quad 4 \\
 & \quad \quad \quad \quad \quad \quad \quad \quad x_1, x_2, x_3, w_4, w_5 \geq 0
 \end{aligned}$$

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	w_4	w_5	
z	1	$1 - 2M$	2	$-6M$			$-8M$
w_4		-2	-1	4	1		4
w_5		4	1	2		1	4

Entering basic variable: x_3

Exiting basic variable: w_4

Example

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	w_4	w_5	
z	1	$1 - 5M$	$2 - 1.5M$		$1.5M$		$-2M$
x_3		-0.5	-0.25	1	0.25		1
w_5		5	1.5		-0.5	1	2



Entering basic variable: x_1
 Exiting basic variable: w_5



Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	w_4	w_5	
z	1		1.7		$M + 0.1$	$M - 0.2$	-0.4
x_3			-0.1	1	0.2	0.1	1.2
x_1		1	0.3		-0.1	0.2	0.4

Example

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	w_4	w_5	
z	1		1.7		$M + 0.1$	$M - 0.2$	-0.4
x_3			-0.1	1	0.2	0.1	1.2
x_1		1	0.3		-0.1	0.2	0.4

The solution is optimal for the model with the artificial variables.

Solution: $(0.4, 0, 1.2, 0, 0)$

Both artificial slack variables are non-basic variables!

The solution $(0.4, 0, 1.2)$ is feasible for the original problem!

Is $x_1 = 0.4$, $x_2 = 0$, $x_3 = 1.2$

an optimal solution for the original problem?

Observations

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	w_4	w_5	
z	1		1.7		$M + 0.1$	$M - 0.2$	-0.4
x_3			-0.1	1	0.2	0.1	1.2
x_1		1	0.3		-0.1	0.2	0.4

If we remove the variables w_4 and w_5 from the tableau, we obtain the tableau of the original problem with the base $\{x_1, x_3\}$!

Basic variable	Coefficient				Solution
	z	x_1	x_2	x_3	
z	1		1.7		-0.4
x_3			-0.1	1	1.2
x_1		1	0.3		0.4

Big-M method: Summary

If all artificial slack variables of the new formulation are 0, then the optimal solution corresponds to the optimal solution of the original formulation.

The additional slack variables can also be added to the previous examples and lead to the same result.

Summary

Simplex algorithm for solving linear optimization problems

- Setting up adjacent BFSs until the optimum is found

Finding the first BFS

- 2-phase simplex
- Big-M method